Rigorous Resolution Refutation and Construction

Part (a): Resolution Refutation Tree

Problem 1. Let φ be the CNF formula defined by the following set of clauses:

$$\begin{aligned}
C_1 &= (x_1 \vee x_2 \vee x_3 \vee x_4) \\
C_2 &= (\neg x_1 \vee x_2 \vee x_4) \\
C_3 &= (x_1 \vee \neg x_2 \vee \neg x_3) \\
C_4 &= (\neg x_2 \vee x_3) \\
C_5 &= (x_2 \vee \neg x_3) \\
C_6 &= (\neg x_4 \vee x_3) \\
C_7 &= (\neg x_1 \vee \neg x_3)
\end{aligned}$$

Show that φ is unsatisfiable using only the resolution rule.

Solution 1. We perform a proof by resolution refutation. We aim to derive the empty clause ($\square$) by resolving pairs of clauses containing complementary literals.

Step-by-Step Derivation

Step 1: Eliminate variable x_4
Resolve C_1 with C_6 on x_4:

$$\frac{(x_1 \vee x_2 \vee x_3 \vee x_4) \quad (\neg x_4 \vee x_3)}{(x_1 \vee x_2 \vee x_3)} \quad \text{(Label } R_1\text{)}$$

Resolve C_2 with C_6 on x_4:

$$\frac{(\neg x_1 \vee x_2 \vee x_4) \quad (\neg x_4 \vee x_3)}{(\neg x_1 \vee x_2 \vee x_3)} \quad \text{(Label } R_2\text{)}$$

Step 2: Eliminate variable x_1
Resolve R_1 with R_2 on x_1:

$$\frac{(x_1 \vee x_2 \vee x_3) \quad (\neg x_1 \vee x_2 \vee x_3)}{(x_2 \vee x_3)} \quad \text{(Label } R_3\text{)}$$

Resolve C_3 with C_7 on x_1:

$$\frac{(x_1 \vee \neg x_2 \vee \neg x_3) \quad (\neg x_1 \vee \neg x_3)}{(\neg x_2 \vee \neg x_3)} \quad \text{(Label } R_4\text{)}$$

Step 3: Eliminate variable x_2
Resolve R_3 with C_4 on x_2:

$$\frac{(x_2 \vee x_3) \quad (\neg x_2 \vee x_3)}{(x_3)} \quad \text{(Label } R_5\text{)}$$

Resolve R_4 with C_5 on x_2:

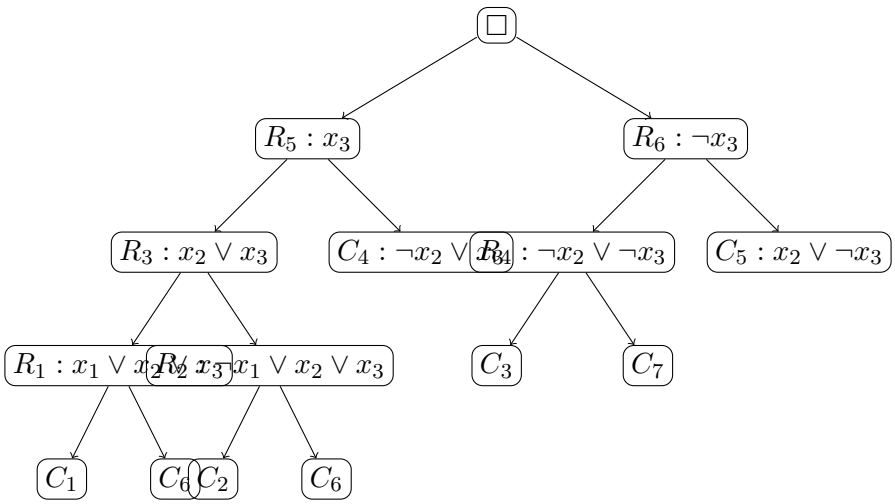
$$\frac{(\neg x_2 \vee \neg x_3) \quad (x_2 \vee \neg x_3)}{(\neg x_3)} \quad \text{(Label } R_6\text{)}$$

Step 4: Derive Empty Clause
Resolve the unit clauses R_5 and R_6 on x_3:

$$\frac{(x_3) \quad (\neg x_3)}{\square}$$

Since we have derived the empty clause, the original formula φ is unsatisfiable.

Resolution Refutation Tree



Part (b): Construction of Unsatisfiable Formula β

Problem 2. Let α be an *unsatisfiable* CNF formula with m clauses over n variables, where the minimum number of literals in a clause is r $(m, n \geq 1, r \geq 2)$. Show that there exists a CNF formula β such that:

- β is unsatisfiable.
- β has $m+1$ clauses and $n+1$ variables.
- α and β share $m-1$ clauses.
- The minimum number of literals in a clause of β is r.

Solution 2. Let the clause set of α be $\{C_1, C_2, \ldots, C_m\}$ over the variable set $X = \{x_1, \ldots, x_n\}$. Since α is unsatisfiable, the conjunction $\bigwedge_{i=1}^{m} C_i$ evaluates to False for all truth assignments to X.

Construction of β

We introduce a new variable $x_{n+1} \notin X$. We select the last clause of α, denoted C_m, and "split" it using the new variable. We define β as the conjunction of the first $m-1$ clauses of α, plus two new clauses derived from C_m:

$$\beta = \left(\bigwedge_{i=1}^{m-1} C_i \right) \wedge (C_m \vee x_{n+1}) \wedge (C_m \vee \neg x_{n+1})$$

The set of clauses for β is:

$$S_\beta = \{C_1, C_2, \ldots, C_{m-1}, D_1, D_2\}$$

where $D_1 = (C_m \vee x_{n+1})$ and $D_2 = (C_m \vee \neg x_{n+1})$.

Verification of Properties

1. β is unsatisfiable:
Suppose for contradiction that β is satisfiable. Then there exists a truth assignment V to variables $\{x_1, \ldots, x_{n+1}\}$ such that all clauses in S_β are true. Specifically, V must satisfy D_1 and D_2.

- If $V(x_{n+1}) = \text{False}$, then $D_1 = (C_m \vee \text{False}) \equiv C_m$ must be true.
- If $V(x_{n+1}) = \text{True}$, then $D_2 = (C_m \vee \text{False}) \equiv C_m$ must be true.

In either case, V must satisfy C_m. Since V also satisfies $\{C_1, \ldots, C_{m-1}\}$, V satisfies all clauses of α. This contradicts the premise that α is unsatisfiable. Thus, β is unsatisfiable.

2. β has $m+1$ clauses and $n+1$ variables:

- **Clauses:** β contains the first $m-1$ clauses of α and the 2 new clauses D_1, D_2. Total clauses $= (m-1) + 2 = m+1$.
- **Variables:** β is defined over $X \cup \{x_{n+1}\}$. Total variables $= n+1$.

3. α and β share $m-1$ clauses:
The intersection of the clause sets is $\{C_1, \ldots, C_{m-1}\}$. The size of this set is exactly $m-1$.

4. Minimum number of literals is r:
Let $|C|$ denote the number of literals in clause C. We are given that $\min_{C \in \alpha} |C| = r$.

- For the shared clauses C_i $(1 \leq i \leq m-1)$, $|C_i| \geq r$.
- For the new clauses D_1 and D_2, $|D_1| = |D_2| = |C_m| + 1$.
- Since $|C_m| \geq r$, we have $|D_1| = |D_2| \geq r + 1$.

The minimum length in β is $\min(\min_{i<m} |C_i|, |C_m| + 1)$. Since all $|C_i| \geq r$, the minimum remains r.

Rigorous Derivation of DNF and Tseitin Encoding

Problem Statement

Consider the formula:

$$\varphi \equiv x_1 \vee \neg(x_2 \wedge \neg(x_3 \leftrightarrow (x_1 \rightarrow \neg x_2)))$$

(a) Give a DNF formula that is semantically equivalent to φ, using techniques studied in class.

(b) Use Tseitin encoding to write an equisatisfiable formula ψ in conjunctive normal form. Would $\neg\psi$ be also equisatisfiable with $\neg\varphi$? Give justification for your answer.

Part (a): Disjunctive Normal Form (DNF)

Solution:

We derive the Disjunctive Normal Form (DNF) by systematically applying logical equivalences.

Step 1: Eliminate Implications and Equivalences

First, we address the innermost implication $x_1 \rightarrow \neg x_2$. Using the equivalence $A \rightarrow B \equiv \neg A \vee B$:

$$(x_1 \rightarrow \neg x_2) \equiv (\neg x_1 \vee \neg x_2)$$

Substituting this back into φ:

$$\varphi \equiv x_1 \vee \neg(x_2 \wedge \neg(x_3 \leftrightarrow (\neg x_1 \vee \neg x_2)))$$

Next, we address the equivalence $A \leftrightarrow B \equiv (A \wedge B) \vee (\neg A \wedge \neg B)$. Let $Q = (\neg x_1 \vee \neg x_2)$.

$$x_3 \leftrightarrow Q \equiv (x_3 \wedge Q) \vee (\neg x_3 \wedge \neg Q)$$

Using De Morgan's Law, $\neg Q = \neg(\neg x_1 \vee \neg x_2) = (x_1 \wedge x_2)$. Thus the equivalence E becomes:

$$E \equiv (x_3 \wedge (\neg x_1 \vee \neg x_2)) \vee (\neg x_3 \wedge (x_1 \wedge x_2))$$

Step 2: Push Negations Inward

The formula is now $\varphi \equiv x_1 \vee \neg(x_2 \wedge \neg E)$. By De Morgan's Law:

$$\varphi \equiv x_1 \vee (\neg x_2 \vee \neg\neg E)$$

By Double Negation Elimination:

$$\varphi \equiv x_1 \vee \neg x_2 \vee E$$

Substituting E back in:

$$\varphi \equiv x_1 \vee \neg x_2 \vee [(x_3 \wedge (\neg x_1 \vee \neg x_2)) \vee (\neg x_3 \wedge x_1 \wedge x_2)]$$

Step 3: Distribute to obtain DNF

We apply the distributive law to $(x_3 \wedge (\neg x_1 \vee \neg x_2))$:

$$(x_3 \wedge \neg x_1) \vee (x_3 \wedge \neg x_2)$$

Substituting this back:

$$\varphi \equiv x_1 \vee \neg x_2 \vee (x_3 \wedge \neg x_1) \vee (x_3 \wedge \neg x_2) \vee (\neg x_3 \wedge x_1 \wedge x_2)$$

Step 4: Simplification

Using the Absorption Law $A \vee (A \wedge B) \equiv A$:
Term 1: x_1 absorbs $(\neg x_3 \wedge x_1 \wedge x_2)$.
Term 2: $\neg x_2$ absorbs $(x_3 \wedge \neg x_2)$.
Reduced DNF:

$$\varphi \equiv x_1 \vee \neg x_2 \vee (x_3 \wedge \neg x_1)$$

Using $A \vee (\neg A \wedge B) \equiv A \vee B$:

$$x_1 \vee (\neg x_1 \wedge x_3) \equiv x_1 \vee x_3$$

Final DNF:

$$\varphi_{\text{DNF}} \equiv x_1 \vee \neg x_2 \vee x_3$$

Part (b): Tseitin Encoding

Solution:

1. Tseitin Construction

We introduce fresh variables y_i for sub-expressions.

$$
\begin{aligned}
&y_1 \leftrightarrow \neg x_2 \\
&y_2 \leftrightarrow (x_1 \rightarrow y_1) \\
&y_3 \leftrightarrow (x_3 \leftrightarrow y_2) \\
&y_4 \leftrightarrow \neg y_3 \\
&y_5 \leftrightarrow (x_2 \wedge y_4) \\
&y_6 \leftrightarrow \neg y_5 \\
&y_7 \leftrightarrow (x_1 \vee y_6)
\end{aligned}
$$

2. CNF Clauses for Definitions

1. Definition of y_1: $(y_1 \leftrightarrow \neg x_2)$

$$(\neg y_1 \vee \neg x_2) \wedge (x_2 \vee y_1)$$

2. Definition of y_2: $(y_2 \leftrightarrow (\neg x_1 \vee y_1))$

$$(\neg y_2 \vee \neg x_1 \vee y_1) \wedge (x_1 \vee y_2) \wedge (\neg y_1 \vee y_2)$$

3. Definition of y_3: $(y_3 \leftrightarrow (x_3 \leftrightarrow y_2))$

$$
\begin{aligned}
&(\neg y_3 \vee \neg x_3 \vee y_2) \wedge (\neg y_3 \vee x_3 \vee \neg y_2) \\
&\qquad \wedge (y_3 \vee x_3 \vee y_2) \wedge (y_3 \vee \neg x_3 \vee \neg y_2)
\end{aligned}
$$

4. Definition of y_4: $(y_4 \leftrightarrow \neg y_3)$

$$(\neg y_4 \vee \neg y_3) \wedge (y_3 \vee y_4)$$

5. Definition of y_5: $(y_5 \leftrightarrow (x_2 \wedge y_4))$

$$(\neg y_5 \vee x_2) \wedge (\neg y_5 \vee y_4) \wedge (\neg x_2 \vee \neg y_4 \vee y_5)$$

6. Definition of y_6: $(y_6 \leftrightarrow \neg y_5)$

$$(\neg y_6 \vee \neg y_5) \wedge (y_5 \vee y_6)$$

7. Definition of y_7: $(y_7 \leftrightarrow (x_1 \vee y_6))$

$$(\neg y_7 \vee x_1 \vee y_6) \wedge (\neg x_1 \vee y_7) \wedge (\neg y_6 \vee y_7)$$

8. Root Unit Clause:

$$(y_7)$$

3. Equisatisfiability of Negations

Question: Is $\neg\psi$ equisatisfiable with $\neg\varphi$?
Answer: No.

Justification: Two formulas A and B are equisatisfiable if (A is satisfiable $\iff$ B is satisfiable).
Case 1: $\neg\varphi$ is satisfiable only if φ is **not** a tautology.
Case 2: $\neg\psi$ is equivalent to $\neg y_{\text{root}} \vee \neg(\text{Definitions})$. Since it is always possible to violate the definitions (by choosing variable assignments that make the gates False), $\neg\psi$ is **always satisfiable**.

Because $\neg\psi$ is always satisfiable regardless of the logical validity of φ, they are not equisatisfiable.

Rigorous Proofs for the Parity Function CNF Properties

Problem Statement

Consider the parity function PARITY : $\{0,1\}^n \to \{0,1\}$, where PARITY$(x_1, \ldots, x_n) = 1$ if and only if an odd number of inputs are 1. We consider a Conjunctive Normal Form (CNF) representation of this function. We assume:

1. No clause contains any variable more than once (e.g., no $p \wedge \neg p$ or $p \vee p$).
2. All clauses are distinct.

(a) Prove that any CNF representation of PARITY must have n literals (from distinct variables) in every clause.
(b) Prove that any CNF representation of PARITY must have $\geq 2^{n-1}$ clauses.

Part (a): Clause Size

Theorem 1. Let ϕ be a CNF formula such that $\phi(x) \equiv \text{PARITY}(x)$ for all $x \in \{0,1\}^n$. Every clause C in ϕ must contain exactly n literals.

Proof. Let $\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$. Since ϕ represents the PARITY function, $\phi(x) = 1$ when the weight of x (number of 1s) is odd, and $\phi(x) = 0$ when the weight of x is even.

For a CNF formula to evaluate to 0 (False) on a specific input x, at least one of its clauses must evaluate to 0. Conversely, if $\phi(x) = 0$, there exists at least one clause $C \in \phi$ such that $C(x) = 0$.

We proceed by contradiction. Assume there exists a clause $C \in \phi$ that has fewer than n literals.

1. **Existence of a Missing Variable:** Since the number of variables is n and $|C| < n$, there exists some variable x_i that does not appear in clause C (neither as a positive literal x_i nor as a negative literal $\neg x_i$).

2. **Falsifying Assignment:** For the clause C to contribute to the function, it cannot be a tautology (always True). In standard CNF form over distinct variables, a clause is never a tautology unless it is empty (False) or contains complementary literals (excluded by problem statement). A clause C evaluates to 0 for exactly those assignments where every literal in C evaluates to 0. Since C is part of ϕ, there must be some input assignment $\alpha \in \{0,1\}^n$ such that $C(\alpha) = 0$. (If C were never 0, it would be always 1, and could be removed without changing the function).

   Since $C(\alpha) = 0$, and ϕ is a conjunction containing C, it follows that $\phi(\alpha) = 0$. By the definition of ϕ, this implies $\text{PARITY}(\alpha) = 0$. Thus, α has an **even** number of 1s.

3. **Flipping the Missing Variable:** Consider a new assignment α' which is identical to α except for the bit at index i:

$$\alpha'_j = \begin{cases} \alpha_j & \text{if } j \neq i \\ 1 - \alpha_j & \text{if } j = i \end{cases}$$

   The weight of α' differs from the weight of α by exactly 1. Since α has even parity, α' must have **odd** parity. Therefore, $\text{PARITY}(\alpha') = 1$. Consequently, for ϕ to be correct, we must have $\phi(\alpha') = 1$.

4. **Contradiction:** Recall that clause C does not contain variable x_i. The truth value of C depends only on the variables present in C. Since α and α' agree on all variables except x_i, and $x_i \notin C$, we must have:

$$C(\alpha') = C(\alpha)$$

   We established earlier that $C(\alpha) = 0$. Therefore, $C(\alpha') = 0$.

   Since ϕ is a conjunction of clauses including C, if $C(\alpha') = 0$, then the entire conjunction $\phi(\alpha')$ must be 0.

   We have derived:

   - $\phi(\alpha') = 1$ (from the definition of PARITY and the bit flip).

- $\phi(\alpha') = 0$ (from the independence of clause C from variable x_i).

This is a contradiction.

Thus, the assumption that $|C| < n$ must be false. Every clause must contain all n variables. $\square$

Part (b): Number of Clauses

Theorem 2. Any CNF representation ϕ of the PARITY function on n variables must contain at least 2^{n-1} clauses.

Proof. Let $\phi = \bigwedge_{j=1}^{m} C_j$. We aim to show $m \geq 2^{n-1}$.

1. **The Set of Falsifying Inputs:** The function PARITY(x) evaluates to 0 if and only if x has an even number of 1s. Let Z be the set of inputs for which the function is False:
$$Z = \{x \in \{0,1\}^n \mid \text{PARITY}(x) = 0\}$$
The number of subsets of $\{1, \ldots, n\}$ with even cardinality is exactly 2^{n-1}. Thus, $|Z| = 2^{n-1}$.

2. **CNF Semantics:** A CNF formula $\phi(x)$ evaluates to 0 if and only if there exists at least one clause C_j such that $C_j(x) = 0$. Therefore, the set of inputs Z where ϕ is False is exactly the union of the sets of inputs falsified by each clause. Let $S_j = \{x \in \{0,1\}^n \mid C_j(x) = 0\}$. We must have:
$$Z = \bigcup_{j=1}^{m} S_j$$

3. **Size of S_j:** From Part (a), we proved that every clause C_j in a CNF for PARITY must contain exactly n literals. A clause with n distinct literals is falsified by exactly one unique input assignment.

   For example, if $n = 3$ and $C = (x_1 \vee \neg x_2 \vee x_3)$, C is False only when $x_1 = 0, x_2 = 1, x_3 = 0$. Any other assignment satisfies at least one literal.

   Therefore, for every clause C_j, we have $|S_j| = 1$.

4. **Counting Argument:** Using the union bound (or simply the sum of cardinalities since sets are disjoint or not), the size of the union cannot exceed the sum of the sizes of the components:
$$|Z| = \left| \bigcup_{j=1}^{m} S_j \right| \leq \sum_{j=1}^{m} |S_j|$$
Substituting the known values:

   - $|Z| = 2^{n-1}$
   - $|S_j| = 1$ for all j

$$2^{n-1} \leq \sum_{j=1}^{m} 1$$

$$2^{n-1} \leq m$$

Therefore, the number of clauses m must be at least 2^{n-1}. $\square$